fully established. In *Tomiopsis* the dorsal median septum seems to arise at the base of the laminated myophore face of the cardinal process[13] as in the Yukon species. In New Zealand, Indian and Australian *Ambikella*, the septum commences a little in front.

Another genus represented in the Yukon is *Attenuatella* Stehli. Although first named from Texas, it is extremely rare in this region (personal communication from G. A. Cooper, Smithsonian Institution), and especially typical of Novaya Zemlya and Taimyr[14] and New Zealand[9]. It also occurs in Peru[15], New Caledonia[16] and to a limited extent in eastern Australia[16].

Whatever the exact generic identities may be, we must conclude that elements of the Canadian brachiopod faunas compare closely with Australian cold-water forms. Canada is not unique among northern countries in this respect. The home of the Licharewiinae and *Tomiopsis* is the eastern Russian Platform and north-east Siberia, and forms possibly akin to the Canadian *Wyndhamia*-like shells have been described from Taimyr[14]. Furthermore, some of the peculiar Australian bivalve genera seem to be represented in north-east Siberia[17–19], as pointed out by Dickins[20]. There thus appears to be evidence of a bipolar fauna, with cool or cold-water elements in both Australasia and the Canadian–Siberian Arctic. A latitude as low as 20° N. for the Canadian faunas seems unlikely, palaeomagnetic evidence notwithstanding. In overall appearances the faunas show some similarities to those of northern Queensland, and especially New Zealand. Judging from the distribution of faunas and glacial sediments, Queensland possibly occupied palaeolatitudes of about 70° to 55° S. in the Permian. This accords well with palaeomagnetic data[21], except that faunas and sediments suggest that only eastern Australia suffered a mid-Permian (Baigendzinian) glaciation, so that the South Pole may have been to the east of the Great Australian Bight, not in the middle as shown by Irving. No palaeomagnetic data are available for New Zealand. The faunas and floras and sediments from this country indicate palaeolatitudes only 10° or so from its present position, perhaps between 55° and 45° S. A slightly lower position would seem acceptable for the Yukon, or even much the same palaeolatitude, if allowance is made for the greater extent of ice in the southern hemisphere. This does not accord well with the palaeolatitude assigned by Irving[3], but agrees more closely with palaeolatitudes for north-east Siberia (ref. 3, Fig. 9.9, page 197). While such disagreement may indicate the crudities and risks involved in trying to calibrate palaeolatitudes through faunal studies, it seems feasible to propose that neither Siberia nor northern Canada has moved in phase with the regions where palaeomagnetic work used by Irving was carried out. The tectonic and stratigraphic maps of the Soviet Union suggest buckling of geosynclines on a huge scale, so that if that part of the continent has moved, it has not moved as an undistorted plate, but has crumpled severely. It is certainly difficult to accept palaeomagnetic reconstructions showing the Texas Permian at 15° S. in a palaeolatitude almost comparable with that of the Yukon and Canadian Arctic at 20° N. Dunbar[22] also emphasized the considerable difference between his Greenland faunas, supposedly with a palaeolatitude close to 20° N., and those of Texas.

In conclusion, it seems from the nature of faunas that the seas covering northern Canada and north-east Siberia in the Permian occupied temperate rather than sub-tropical palaeolatitudes, high enough to support a number of cool or cold-water loving genera, at least from time to time, presumably when glacial spasms gripped the southern hemisphere.

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Palaeomagnetic Evidence for the Ordovician Geomagnetic Pole Position

I HAVE obtained evidence which supports a pole position in the region of 165° E., 13° N. (antipole 15° W., 13° S.) on the present latitude–longitude grid and relative to the British Isles. This agrees with the palaeoclimatological evidence reported by Spjeldnaes[1], whose studies, based chiefly on the global distribution of shallow water shell deposits, led him to suggest that the Ordovician climatic zones could be deduced from the faunal provinces and that polar ice caps existed during this period. Spjeldnaes[1] infers a rotational pole lying in or west of Africa which would correspond to the palaeomagnetic pole at 15° W., 13° S. These data are not in conflict with Wegener's hypothesis of continental drift, for the positions of the climatic zones on either side of the Atlantic are improved by moving the continental blocks together[1]. There is little corresponding information for the 165° E., 13° N. pole because the data available for China are too few to say whether the palaeolatitude and palaeomagnetic latitude are in agreement. Spjeldnaes concluded from a consideration of Runcorn's[2] palaeomagnetic polar wandering curves that a magnetic pole at about 165° E., not far from the Equator, in the region of 0°–25° north of it, would correlate well with the available data.

Oriented hand samples were collected from seven lava flows in the Eycott group in the Borrowdale volcanic series of Llanvirn–Llandeilo age. Cylinders were cored out of these in the laboratory and an astatic magnetometer was used to measure the remanence of the cores.

Table 1

Eycott	Long.	Lat.	α	Circular standard deviation	Circular standard error	No. of samples	R	Polarity
(1) (2° 56′ W., 54° 40′ N.)	165° E.	**14° N.**	11·4°	15·1°	5·3°	7	**6·794**	N.
(2) Llanelwedd (3° 24′ W., 52° 9′ N.)	162° E.	**15° N.**	13·9°	21·0°	7·0°	9	**8·453**	S.
(3) Stockenray Bay (4° 54′ W., 55° 11·5′ N.)	168° E.	**11° N.**	9·5°	17·4°	5·0°	12	**11·495**	**S.**

(Note: Fisherian[5] statistics used R, α.)

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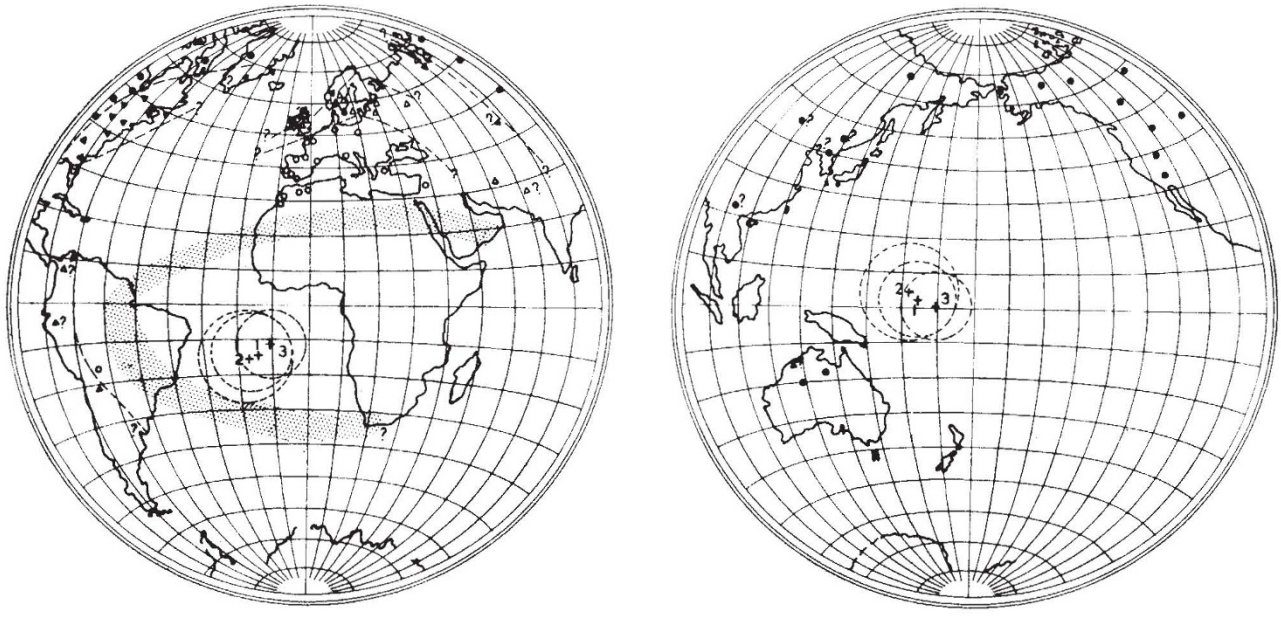


Fig. 1. Comparison of the proposed pole position with Spjeldnaes' climatic data. ●, Tropical faunas; ▲, warm temperate faunas; △, cold temperate faunas; ○, Arctic and Antarctic type faunas. Numbers 1, 2 and 3 refer to the sites in Table 1.

These samples were then subjected to a.c. and thermal demagnetization tests. On the basis of these tests three hand samples were rejected in calculating a mean direction of magnetization. Two of these included the Earth's present field direction within the circular standard deviation of their field directions at the site. The third was strongly magnetized in a direction remote from the remaining samples, but its mean direction began to migrate towards the main group on a.c. cleaning in fields above 560 peak oersteds, moving into the same quadrant after 900 peak oersteds. The specimens initially had a geometric mean intensity $58{\cdot}7 \times 10^{-4}$ (*S.D.* logs 0·37) gauss, susceptibility of $16{\cdot}0 \times 10^{-3}$ (*S.D.* logs 0·42) gauss/oersted, and intensity to susceptibility ratio of 0·37 (*S.D.* logs 0·37) oersted. A pole position corresponding to the mean direction of magnetization (after cleaning in 85 peak oersteds) was calculated for each flow and a mean pole (Table 1) for the site calculated from these (Fig. 1 (1)).

Evidence supporting this direction was obtained from hand samples of Upper Llanvirn[3] age from the Builth volcanic series collected at Llanelwedd in the Builth Wells–Llandrindod inlier. Samples were subjected to a.c. and thermal demagnetization tests on the basis of which three hand samples out of twelve had to be rejected. The mean directions from the other hand samples were combined to yield the mean pole position (Fig. 1 (2) and Table 1 (2)). Most of these samples were very weak with geometric mean intensities in the range $1{\cdot}2–20{\cdot}7 \times 10^{-6}$ gauss and susceptibilities in the range $(12{\cdot}7–52{\cdot}0) \times 10^{-6}$ gauss/oersted.

Twelve samples were also collected from four sites in the Arenig lavas in Stockenray Bay south of Girvan. These had a geometric mean intensity of $75{\cdot}0 \times 10^{-6}$ gauss (*S.D.* logs 0·11), susceptibility of 119×10^{-6} gauss/oersted (*S.D.* logs 0·18), and intensity/susceptibility ratio of 0·75 (*S.D.* logs 0·12) oersted. It was found that a.c. demagnetization reduced the scatter between samples from the same site and the scatter of the directions between sites, particularly when these directions were referred to the horizontal plane rather than the local bedding direction. This demonstrated that the magnetic age probably corresponds to the age of the folding of these rocks which occurred before the deposition of the Bala sediments which overstep the Arenig rocks of the Girvan[4] region. This could therefore be an Ordovician direction.

Table 2. THE MAGNETIZATION OF SAMPLES FROM STOCKENRAY BAY

	Dec.	Inc.	No. of samples	R	Circular standard deviation	Circular standard error
Horizontal plane	184°	54°	12	10·674	28°	8°
Bedding plane	202°	32°	12	8·621	46°	·. 13°
After 85 oersteds						
Horizontal plane	189°	40°	12	11·511	17°	5°
Bedding plane	202°	21°	12	9·216	42°	12°

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PHYSICS

Growth and Destruction of Ice Filaments in an Electric Field

MODIFICATION by electric fields of the growth of ice crystals has been observed under various conditions. Schaefer[1], who was the first to report rapid growth of ice in the form of whisker-like aggregates at high electrical gradients, observed that super-cooled water droplets were projected outwards from the whisker tips. Marshall and Gunn[2] found that the growth of ice crystals in relatively weak electric fields was initiated as opaque sprouts, which later grew many irregular branches of small cross-section and of no recognizable crystal habit. More recent observations[3,4], including studies of the growth of ice crystals in a diffusion cloud chamber, have yielded essentially similar results.

There are three chief factors which indicate the importance of studying the effect of electric fields on ice crystal growth from the vapour phase. First, electric

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